

Same Data, Different Bins

AP Statistics · Topic 1.5 (CED 1.5) · B: 2026-09-15 · E: 2026-09-18 · TEACHER KEY

CED TETHER

- **Skill 2.A** — Describe data presented numerically or graphically.
- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **EU UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **UNC-1.F** Classify types of quantitative variables. [Skill 2.A]
 - **UNC-1.F.1** A discrete variable can take on a countable number of values. The number of values may be finite or countably infinite, as with the counting numbers. **UNC-1.F.2** A continuous variable can take on infinitely many values, but those values cannot be counted. No matter how small the interval between two values of a continuous variable, it is always possible to determine another value between them.
- **UNC-1.G** Represent quantitative data graphically. [Skill 2.B]
 - **UNC-1.G.1** In a histogram, the height of each bar shows the number or proportion of observations that fall within the interval corresponding to that bar. Altering the interval widths can change the appearance of the histogram.
 - **UNC-1.G.2** In a stem and leaf plot, each data value is split into a "stem" (the first digit or digits) and a "leaf" (usually the last digit).
 - **UNC-1.G.3** A dotplot represents each observation by a dot, with the position on the horizontal axis corresponding to the data value of that observation, with nearly identical values stacked on top of each other.
 - **UNC-1.G.4** A cumulative graph represents the number or proportion of a data set less than or equal to a given number.
 - **UNC-1.G.5** There are many additional ways to graphically represent distributions of quantitative data.

Essential question: How can the same quantitative data tell a different visual story when the histogram bins change?

DOK-3 skill: 4.B · **FRQ pattern:** bin-width-which-display-answers-the-question-and-what-it-hides

DOK rationale: Part (c) weighs two bin widths against the coach's purpose, identifies structure lost through grouping, bounds an unlabeled-group conclusion, and defends when the coarser display is preferable.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Read Priya's and Leo's claims once; do not endorse either.
- Start the video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and press for a graph feature plus a limit on the conclusion.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to F or C and name one visible feature before instruction.
- Complete the video follow-along on quantitative displays and histogram construction.
- Finish (a)–(c), complete the exit line, and turn in the sheet.

Questions to ask

- Which observations changed when the bin width changed?
- Point to the feature Leo can see in F that Priya cannot see in C.
- Where in either histogram is the school-day label stored?
- Name a question for which adding five narrow-bin counts is less useful than reading one broad bar.

Adult role

Solo teacher. Circulate about five pairs. Ask students to separate what the graph suggests from what the unlabeled graph establishes; do not confirm F until both are addressed.

[WARN] WATCH FOR

- Treating the two histograms as different samples rather than two bin choices for the same 30 days.
- Assigning the lower cluster to school days and the upper cluster to weekends without checking date labels.
- Claiming that narrow bins are automatically more honest for every question.

SCORING PART (c) — E / P / I

Expected elements

- **selects-fine-for-purpose** (required) — Chooses F or Leo as the better first look for the coach
- **cites-hidden-feature** (required) — Cites a cluster or the 9–11-thousand gap visible in F and merged in C
- **states-misleading-conclusion** (required) — Explains that C can suggest one routine and hide two possible groups
- **bounds-group-conclusion** (required) — Requires day labels before assigning either cluster to school days or weekends
- **defends-coarse-use** (required) — Names a broad-bin question for which C is more efficient and explains why

E	Chooses F, cites structure C merges, explains the one-routine misread, requires day labels, and defends a concrete use for C.
P	Makes the supported choice but omits the inference limit, misleading consequence, or defensible use for C.
I	Chooses by appearance, treats bin width as changing the data, assigns unlabeled clusters to day types, or cites no feature.

Common mistakes

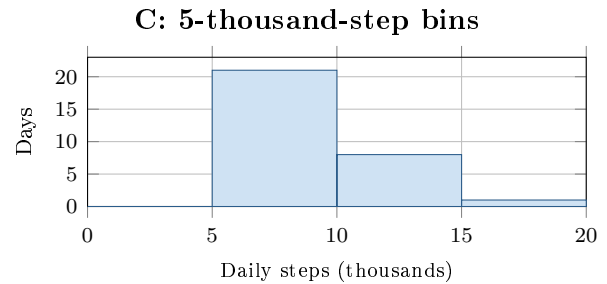
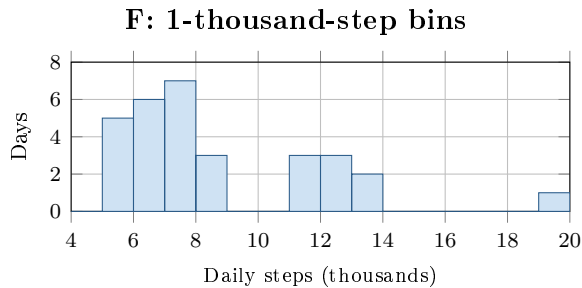
- Saying F contains more observations although both show the same 30 days
- Assigning a cluster to weekends without day labels
- Saying narrower bins are always better instead of matching the question

[STAR] TODAY'S PROBLEM — annotated

Suppose a student recorded steps for 30 days. Histograms F and C show the same data with 1-thousand- and 5-thousand-step bins. A coach asks, “Does this student move differently on school days and weekends?”

Priya: “Use C. F has too many short and empty bars; C gives the clearer summary.”

Leo: “Use F. C combines values that should not be treated as one group.”



FIRST TAKE (5 min)

Which histogram is more useful to the coach, F or C? Name one visible feature.

Key: Not graded. Expect a split between visual simplicity and retained detail.

Rules callout “READING HISTOGRAMS (keep on the page)” is printed on the student sheet.

DOK 1 (a) State each bin width and the number of days in F’s 19–20-thousand interval.

Key: F uses 1-thousand-step bins; C uses 5-thousand-step bins. F has 1 day in 19–20 thousand.

DOK 2 (b) Describe F in context, including clusters, gaps, and the unusual value, with units.

Key: There are 21 days from 5.1–8.6 thousand, 8 from 11.1–13.4 thousand, a gap from 8.6–11.1, a longer gap to an isolated 19.0 thousand. Raw data (thousands): 5.1, 5.3, 5.5, 5.7, 5.8, 6.1, 6.2, 6.4, 6.6, 6.8, 6.9, 7.1, 7.2, 7.4, 7.5, 7.7, 7.8, 7.9, 8.1, 8.3, 8.6, 11.1, 11.4, 11.8, 12.1, 12.4, 12.7, 13.1, 13.4, 19.0.

DOK 3 (c) Adjudicate Priya’s and Leo’s recommendations. Choose the better first look for the coach, cite a feature that vanishes in the other display, and explain what its reader could wrongly conclude. State what information is still needed before naming either cluster school days or weekends. Give one specific question for which C is better.

Key: Leo’s F is the better first look: it preserves clusters near 5–9 and 11–14 thousand and the 9–11 gap. C merges these into adjacent broad bars, so a reader could infer one right-skewed routine and miss two possible routines. Neither histogram identifies day type; match observations to dates or compare labeled school-day/weekend displays. C is better for “How many days were below 10 thousand?” because its broad bin shows 21 directly.

EXIT

One thing the video changed about my first take: _____